Augmented Offense — Module 4: Cryptobionic security (VivoKey)

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Phone-native · ~15 min (1:23–1:38) · the exhale

Module 4 — Cryptobionic security: a credential with nothing to clone

Implant analog: Apex / Spark. **Tool:** your phone + the **Apex Manager** app.

> **Use the app named "Apex Manager"** (by VivoKey) — *not* the generic VivoKey or Fidesmo app. Apex Manager is the one that reads the Apex/VivoKey test card's secure element. It's free on the **Apple App Store** and **Google Play**; have attendees install it during the opening (project the store QR). **This is the defensive capstone** — the room has broken three things; now show the one that holds and why.

Objective attendees reach

Scan the VivoKey test card and see challenge-response / cryptographic identity where the keys **never leave the secure element** — no dump, no UID to forge, no replay.

Materials

- Attendee phone (iPhone or Android) with the **Apex Manager** app installed
- 1× VivoKey test card per attendee (or pass a few around)

Step by step (both platforms)

1. Open the **Apex Manager** app.
2. **Scan** the VivoKey test card (hold it to the top-back of the phone).
3. Watch the cryptographic identity / challenge-response flow.
4. **Contrast out loud against Modules 2–3:** try to “dump” it — there’s nothing to dump. There’s no UID to forge and no replay.

Why it matters

This is the implant you can scan and still never become. Everything you broke earlier assumed the credential was a secret you could read — this one never hands you the secret.

Defensive synthesis

Pull the three broken primitives into adversary workflow and defense:

- **Attacker-as-hardware** collapses proximity and intent assumptions — the operator *is* the tool.
- **Access control:** UID ≠ identity. Retire LF prox and bare MIFARE Classic; move to challenge-response / identity-bound credentials (this card).
- **NFC/BLE policy:** treat tap-to-URL as untrusted input; MDM on background NDEF; behavior training against redirect chains.
- **Detection:** what does defender *visibility* look like when the tool is subdermal? Name the gap — that’s the research contribution (see the Defense guide).
- **Cross-domain:** cyber + physical + cognitive/IO — the DEF CON abstract’s core claim.

Failure modes

Symptom	Cause	Fix
App won’t scan	NFC off / wrong phone position	enable NFC, hold card to the top of the phone
Attendees “want to clone it too”	that’s the point	there’s nothing to clone — that’s the whole lesson

Do not cut

This module resolves the tension of the whole session. If time is tight, cut depth from Module 3, never this.

Safety

Read-only scan. Nothing is written, forged, or captured.

Presenters

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